```
# Task1
text = "Artificial Intelligence"

print(text[:10])

print(text[-5:])

print(text[::2])

print(text[::-1])

print(text[11:])

# Task2
email = "student@gmail.com"

print(email[:7])

print(email[8:13])

print(email[-3:])

# Task3
numbers = [10, 20, 30, 40, 50]

numbers.append(60)
print(numbers)

numbers.remove(30)
print(numbers)

numbers.insert(2, 25)
print(numbers)

maximum = max(numbers)
```

```
minimum = min(numbers)
print("Max:", maximum)
print("Min:", minimum)

numbers.reverse()
print(numbers)

# Task4
data = [12, 45, 67, 23, 89, 45, 12, 90]

data = list(set(data))
print(data)

asc = sorted(data)
print("Ascending:", asc)

desc = sorted(data, reverse=True)
print("Descending:", desc)

# Task5
t = (10, 20, 30, 40, 50)

print(t[0])
print(t[-1])

print(len(t))

print(30 in t)

l = list(t)
print(l)
```

```
# Task6
student = {
    "name": "Rahul",
    "age": 21,
    "marks": 85
}

print(student.keys())

print(student.values())

student["grade"] = "A"
print(student)

student["marks"] = 90
print(student)

student.pop("age")
print(student)

# Task7
def add(a, b):
    return a + b
print("Result:", add(5, 3))

def even_odd(num):
    if num % 2 == 0:
        return "Even"
    else:
        return "Odd"
print("Result:", even_odd(7))

def factorial(n):
    fact = 1
    for i in range(1, n + 1):
```

```
        fact *= i
    return fact
print("Fact:", factorial(5))
```

 # Task8

```
for i in range(1, 6):
    print(str(i) * i)
```

 # Task9

```
num = 12345
total = 0

for digit in str(num):
    total = total + int(digit)

print("Result:", total)
```

 # Task10

```
class BankAccount:

    def __init__(self, account_number, balance):
        self.account_number = account_number
        self.balance = balance

    def deposit(self, amount):
        self.balance = self.balance + amount
        print("Amount deposited:", amount)

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance = self.balance - amount
            print("Amount withdrawn:", amount)
        else:
            print("Insufficient balance")
```

```
    def display_balance(self):
        print("Account Number:", self.account_number)
        print("Current Balance:", self.balance)

acc1 = BankAccount(12345, 1000)

acc1.deposit(500)
acc1.withdraw(300)
acc1.display_balance()
```

 # Task11

```
data = {
    "Name": ["A", "B", "C"],
    "Marks": [85, 90, 78],
    "Age": [20, 21, 19]
}

df = pd.DataFrame(data)

print("First Row:")
print(df.iloc[0])

print("\nMarks Column:")
print(df["Marks"])
```

 # Task12

```
import pandas as pd

df = pd.read_csv("Students.csv")

print("First 5 Rows:")
print(df.head())

print("\nLast 5 Rows:")
print(df.tail())
```

```
print("\nColumn Names:")
print(df.columns)

df_no_null = df.dropna()
print("\nData after removing null values:")
print(df_no_null)

df_filled = df.fillna(df.mean(numeric_only=True))
print("\nData after replacing null values with mean:")
print(df_filled)
```

Outputs:

```
PS D:\Internship\Day19> python task1.py
Artificial
gence
Atfca nelgne
ecnegilletnI laicifitrA
Intelligence
```

```
PS D:\Internship\Day19> python task2.py
student
gmail
com
```

```
PS D:\Internship\Day19> python task3.py
[10, 20, 30, 40, 50, 60]
[10, 20, 40, 50, 60]
[10, 20, 25, 40, 50, 60]
Max: 60
Min: 10
[60, 50, 40, 25, 20, 10]
```

```
PS D:\Internship\Day19> python task4.py
[67, 12, 45, 23, 89, 90]
Ascending: [12, 23, 45, 67, 89, 90]
Descending: [90, 89, 67, 45, 23, 12]
```

```
PS D:\Internship\Day19> python task5.py
10
50
5
True
[10, 20, 30, 40, 50]
```

```
PS D:\Internship\Day19> python task6.py
dict_keys(['name', 'age', 'marks'])
dict_values(['Rahul', 21, 85])
{'name': 'Rahul', 'age': 21, 'marks': 85, 'grade': 'A'}
{'name': 'Rahul', 'age': 21, 'marks': 90, 'grade': 'A'}
{'name': 'Rahul', 'marks': 90, 'grade': 'A'}
```

```
PS D:\Internship\Day19> python task7.py
Result: 8
Result: Odd
Fact: 120
```

```
PS D:\Internship\Day19> python task8.py
1
22
333
4444
55555
```

```
PS D:\Internship\Day19> python task9.py
Result: 15
```

```
PS D:\Internship\Day19> python task10.py
Name: ABC
Age: 21
Marks: 85
Amount deposited: 500
Amount withdrawn: 300
Account Number: 12345
Current Balance: 1200
```

```
PS D:\Internship\Day19> python task11.py
First Row:
Name      A
Marks    85
Age      20
Name: 0, dtype: object

Marks Column:
0    85
1    90
2    78
Name: Marks, dtype: int64
```

```
PS D:\Internship\Day19> python task12.py
First 5 Rows:
   id   name   age marks       city
0   1   Asha  21.0    85     Mysore
1   2   Ravi  22.0    90  Bangalore
2   3   John   NaN    78  Mangalore
3   4  Meena  21.0   abc      Delhi
4   5  Kiran  23.0    88    Chennai

Last 5 Rows:
   id   name   age marks       city
4   5  Kiran  23.0    88    Chennai
5   6   Ravi  22.0    90  Bangalore
6   7  Pooja   NaN    95     Mysore
7   8   Arun  24.0    70  Hyderabad
8   3   John   NaN    78  Mangalore

Column Names:
Index(['id', 'name', 'age', 'marks', 'city'], dtype='object')

Data after removing null values:
   id   name   age marks       city
0   1   Asha  21.0    85     Mysore
1   2   Ravi  22.0    90  Bangalore
3   4  Meena  21.0   abc      Delhi
4   5  Kiran  23.0    88    Chennai
5   6   Ravi  22.0    90  Bangalore
7   8   Arun  24.0    70  Hyderabad

Data after replacing null values with mean:
   id   name        age marks       city
0   1   Asha  21.000000    85     Mysore
1   2   Ravi  22.000000    90  Bangalore
2   3   John  22.166667    78  Mangalore
3   4  Meena  21.000000   abc      Delhi
4   5  Kiran  23.000000    88    Chennai
5   6   Ravi  22.000000    90  Bangalore
6   7  Pooja  22.166667    95     Mysore
7   8   Arun  24.000000    70  Hyderabad
8   3   John  22.166667    78  Mangalore
```